

Metropolitan General Hospital - Division of Hematology
DEATH SUMMARY

DATE OF ADMISSION (Final): July 18, 2023

DATE OF DEATH: August 5, 2023 at 21:30

REASON FOR FINAL ADMISSION: Refractory Acute Myeloid Leukemia with Sepsis.

SUMMARY OF CLINICAL COURSE:

Ms. James was a 66-year-old female with a past medical history of GERD and osteoarthritis who was diagnosed with de novo Acute Myeloid Leukemia in November 2022. Her case was unfortunately characterized by a highly aggressive disease biology from the outset.

Initial Diagnosis and Treatment (November 2022):

Ms. James presented to the emergency department with severe fatigue and gingival bleeding. Her initial workup revealed hyperleukocytosis and pancytopenia. A bone marrow biopsy performed on November 4, 2022, confirmed AML with 75% blasts. The critical finding from her molecular panel was a mutation in the **STAG2 gene**. This mutation is a cohesin complex gene alteration that places a patient into the **ELN 2022 Adverse-risk** category, portending a poor prognosis and resistance to conventional chemotherapy.

Given her age (66) and the adverse-risk biology of her disease, she was not considered an ideal candidate for intensive induction chemotherapy. The decision was made to proceed with the standard-of-care, less-intensive regimen of **Venetoclax combined with Azacitidine**, which was initiated on November 11, 2022.

She had a promising initial response to therapy. A bone marrow biopsy after two cycles showed a significant reduction in blasts to <5%, achieving a morphologic complete remission. For the first five months of her treatment, she enjoyed a markedly improved quality of life, was transfusion-independent, and was able to engage in her daily activities.

Disease Progression and Decline (May 2023):

In May 2023, approximately six months after starting treatment, her routine CBCs began to show a steady decline in her platelet and neutrophil counts. She reported a recurrence of profound fatigue. A bone marrow biopsy was performed and confirmed a morphologic relapse of her AML, with marrow blasts having increased to 40%. Her disease had become refractory to the Venetoclax/Azacitidine regimen.

Final Hospitalization and End-of-Life Care (July - August 2023):

The oncology team had an extensive discussion with Ms. James and her family regarding the extremely poor prognosis of relapsed, adverse-risk AML. We discussed options including enrollment in a Phase I clinical trial or intensive salvage chemotherapy, but it was made clear that the likelihood of a meaningful response was exceptionally low, while the certainty of severe toxicity was high. After careful consideration, Ms. James made the definitive decision to decline further anti-leukemic therapy and to transition to comfort-focused care.

She was admitted to the hospital on July 18, 2023, with neutropenic fever and severe mucositis. Blood cultures grew *E. coli*, and she was started on broad-spectrum antibiotics. While her fever initially improved, she never regained her strength. Her pancytopenia worsened, and she developed significant bleeding requiring daily platelet transfusions. Her

care was formally transitioned to the Palliative Care service. Her pain from mucositis and bone infiltration was managed with a continuous intravenous infusion of hydromorphone.

Over her final two weeks, she became progressively weaker and somnolent but remained comfortable and was able to have meaningful interactions with her family, who were a constant presence. On the evening of August 5, 2023, she experienced an episode of respiratory distress and passed away peacefully, approximately nine months after her initial diagnosis.

CAUSE OF DEATH:

- **Primary:** Sepsis secondary to neutropenia.
- **Underlying Cause:** Refractory Acute Myeloid Leukemia.

PATIENT: JAMES, Cheryl

MRN: AML074

DOB: July 27, 1956